BERNSTEIN
SOCIETE GENERALE GROUP

Weekend Tech Byte: The 'cost' of cards — the often misleading comparisons; our perspectives in key charts

Harshita Rawat, CFA +1 917 344 8485 harshita.rawat@bernsteinsg.com
Stacy A. Rasgon, Ph.D. +1 213 559 5917 stacy.rasgon@bernsteinsg.com
Mark C. Newman +1 212 845 7822 mark.newman@bernsteinsg.com
Mark L. Moerdler, Ph.D. +1 917 344 8506 mark.moerdler@bernsteinsg.com
Peter Weed +1 917 344 8390 peter.weed@bernsteinsg.com
Nikhil Devnani, CFA +1 917 344 8425 nikhil.devnani@bernsteinsg.com
Madison Rezaei +1 917 344 8622 madison.rezaei@bernsteinsg.com
Gautam Chhugani +91 226 842 1416 gautam.chhugani@bernsteinsg.com
Laurent Yoon +1 917 344 8502 laurent.yoon@bernsteinsg.com
Viola Chen +1 917 344 8614 viola.chen@bernsteinsg.com
Simran Ratani +1 917 344 8329 simran.ratani@bernsteinsg.com

By Harshita Rawat

The cost of acceptance of cards comes up quite often in discussions about cards. Many of the comparisons (vs. other payment methods) are often not apples-to-apples and somewhat misleading. Debit cards (often functionally similar to other alternatives) are already quite cheap. Credit cards offer credit and rewards to consumers. There are then services cards wrap around a transaction such as dispute resolution, fraud protection, greater conversion, universal acceptance, etc.

That said, there is some growing merchant push back against credit cards' costs in the U.S. (e.g., via surcharges by some small merchants). Taking a step back, interchange requires a delicate balance and the networks can (somewhat) be flexible with interchange to make certain categories addressable.

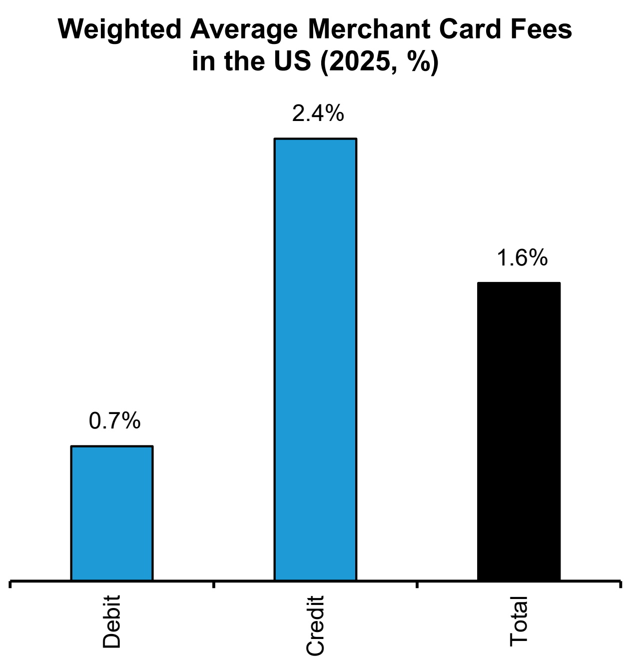
In this short weekend tech byte, we discuss the nuanced topic of cost of acceptance for cards (via our favorite charts) and the implications for cards' growth in the future (including the agentic future).

THE APPLES-TO-ORANGES COMPARISON

First things first: many cost comparisons for cards vs. other payment methods are often misleading. Remember, debit is money you have. Credit is, well, credit (money you don't have). Alternative ways to pay (e.g., pay by bank) are functionally similar to debit, NOT credit. Debit cards are already quite cheap. Credit alternatives (e.g., BNPL) often carry a higher merchant discount rate (MDR) vs. credit cards.

For e.g., in the U.S. regulated debit MDR (merchant rates) averages around 70bps vs. credit at 240bps (Exhibit 1). Note that the unregulated debit (about a third of debit volume in the U.S.) is more expensive (~100bps higher) than regulated debit.

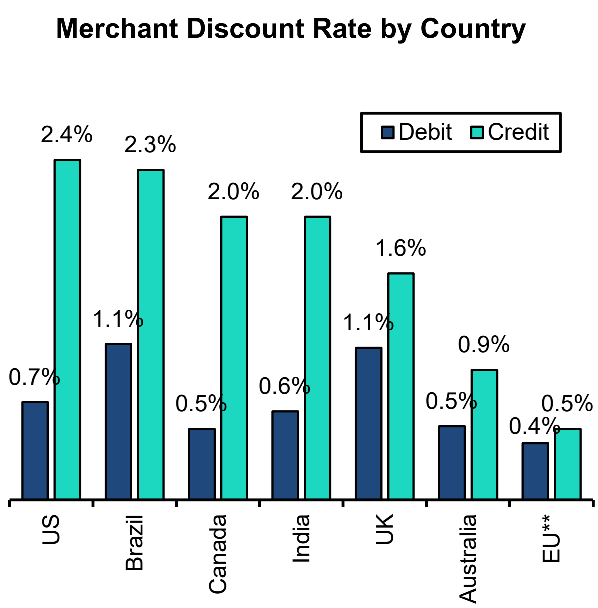
EXHIBIT 1: **Debit is significantly cheaper vs. credit in the U.S.**



Source: Nilson, Bernstein analysis

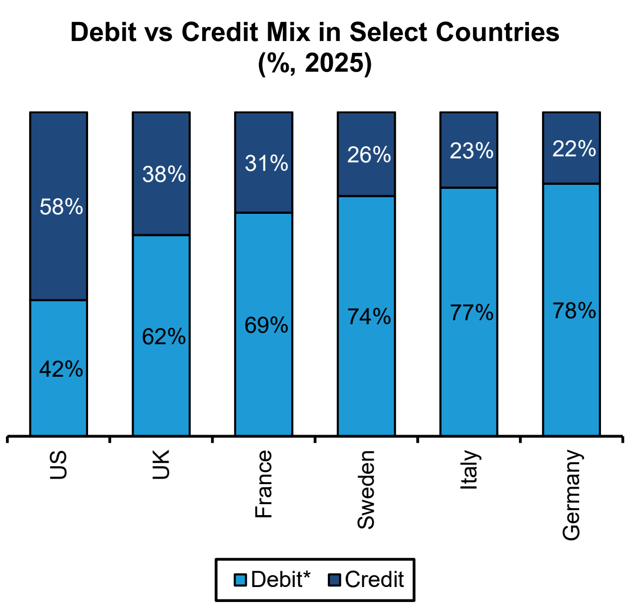
Cards are especially cheap in Europe (Exhibit 2) where interchange is capped at ~20bps for debit and ~30bps for credit. Credit is more expensive than debit in most countries around the world. But, with the exception of the U.S., the majority of card volumes in many geos are debit-related (Exhibit 3).

EXHIBIT 2: **Cards are especially cheaper in Europe**



Source: Central bank data, Bernstein estimates and analysis

EXHIBIT 3: **Mix between credit/debit cards varies widely by country..**



Debit includes Prepaid Cards
Source: Global Payments Report, Bernstein analysis

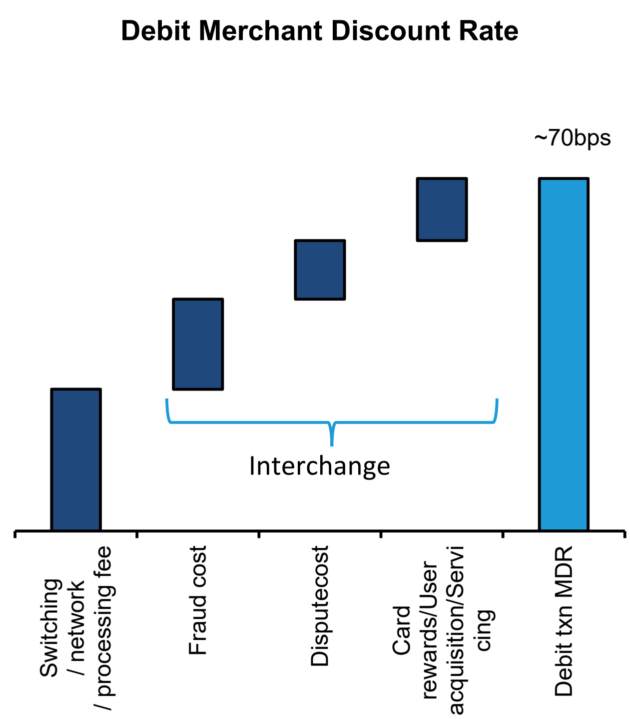
CARDS WRAP AROUND A LOT OF SERVICES

For example, dispute management (doesn't currently exist with Stablecoins), chargebacks (Stablecoin transactions are non-reversible) and rewards (float-income-induced stablecoin rewards have mathematical limits) (Exhibit 4 and Exhibit 5). Can alternative payment methods (pay-by-bank, stablecoins) include these services? Of course, they can - but it takes years and costs $$s (and bragging rights around being 'cheap').

Card payments are beyond simplistic money movement. Money movement is a commodity (ACH transactions cost a fraction of a cent — cheaper than many stablecoins). Trust and governance (which cards provide) are not. There are a lot of value-added services (some described above) embedded in a transaction, and they also include governance standards (which have been decades in the making, and typically very hard to enforce for alternate payment schemes). There is then the universal acceptance across 100m+ merchant locations and 7B+ KYCed credentials.

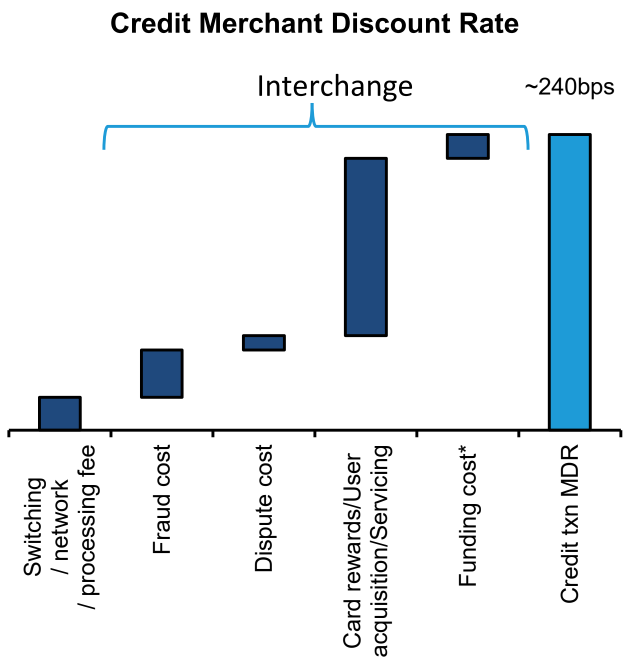
Interestingly, with wrapped services, stablecoins' costs to small merchants are not often cheaper vs. debit cards (Exhibit 6).

EXHIBIT 4: **Cards wrap around a lot of services for debit..**



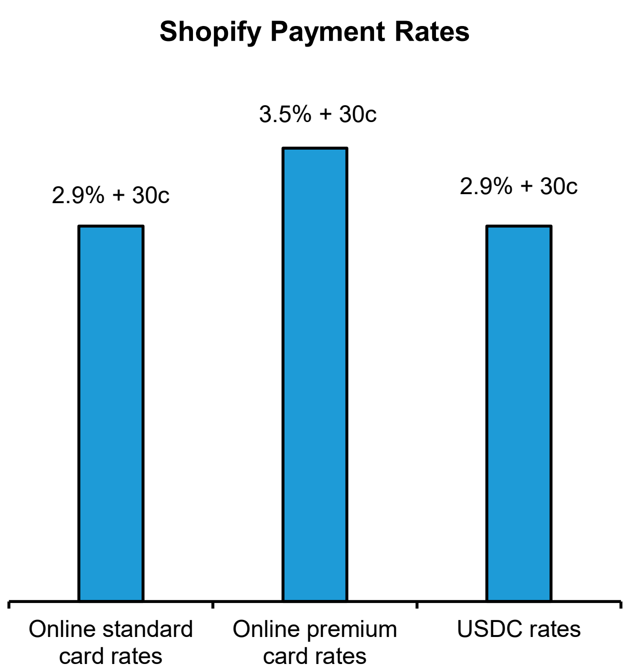
Source: Fed, CFPB, Nilson, Bernstein analysis and estimate

EXHIBIT 5: **..and credit**



Doesn't include losses
Source: Fed, CFPB, Nilson, Bernstein analysis and estimate

EXHIBIT 6: **Stablecoins' costs to small merchants (rack rates) are not often cheaper vs. debit cards**



Source: Corporate websites, Bernstein analysis

INTERCHANGE IS GREAT, BUT CAPS ON IT ARE NOT THAT BAD EITHER..

Interchange is a delicate topic in the payments world. It is one of the many less-than-intuitive things about the sector. It is a fee set by the networks (i.e. Visa and Mastercard), which a merchant's bank pays to the issuing bank for card transactions. Interchange is often invisible to the consumer but passed on to the merchant. It constitutes the majority of fees that merchants pay to accept cards.

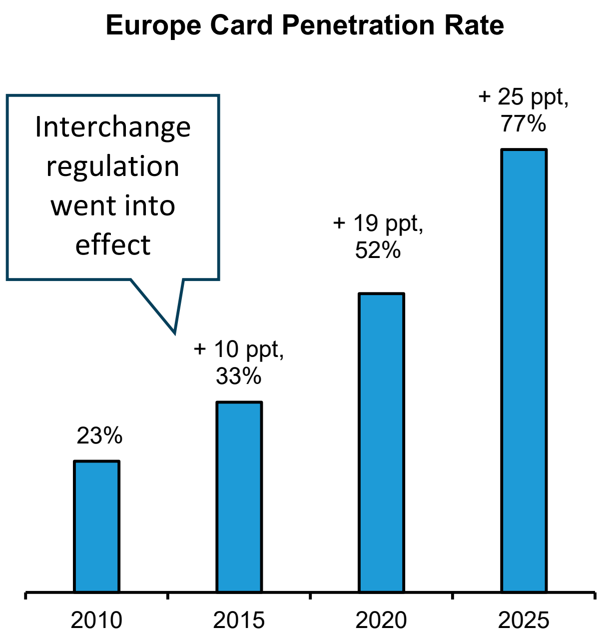
Merchants don't like it, consumers do not care about it, issuers love it, regulators want to cap it, and many lawyers often pay their bills with it. Above all, interchange has, in aggregate, been brilliant. It creates incentives for issuers (including fintechs) to acquire customers and issue cards. It funds rewards, which then incentivize consumer behavior.

Interchange creates a monetization model (beyond lending and insurance) for consumer financial services. The fintechs (new-age issuers), which the so-called disruptors, are now increasingly relying on interchange to drive revenues. Ironically, the best consumer use-case for stablecoins is a stablecoin-linked card, which also benefits from lucrative interchange fees.

All that said, card networks try to achieve a tricky balance for interchange (balancing merchants vs. issuers' interests) — and sometimes regulation isn't all that bad. For e.g., Europe capped interchange fees at 20bps for debit and 30bps for credit in Dec 2015 - and since then card penetration has massively increased in the region (the pandemic-driven digitization also helped here) (Exhibit 7).

Note that the currently **pending MDL settlement in the U.S., if approved, will bring credit card interchange down by ~10bps for 5 years**, and cap standard consumer credit rates for up to eight years. **We are also monitoring the possibility that the Fed lowers regulated debit interchange in the U.S.** The Fed is long overdue (since 2023) on rule-making on its proposed ~30% reduction in effective regulated debit interchange. There have also been some recent conflicting court rulings regarding regulated debit interchange.

EXHIBIT 7: **Europe interchange regulation went into effect in Dec 2015, significantly boosting card penetration and adoption**



Source: Global Payment Reports, Bernstein estimates and analysis

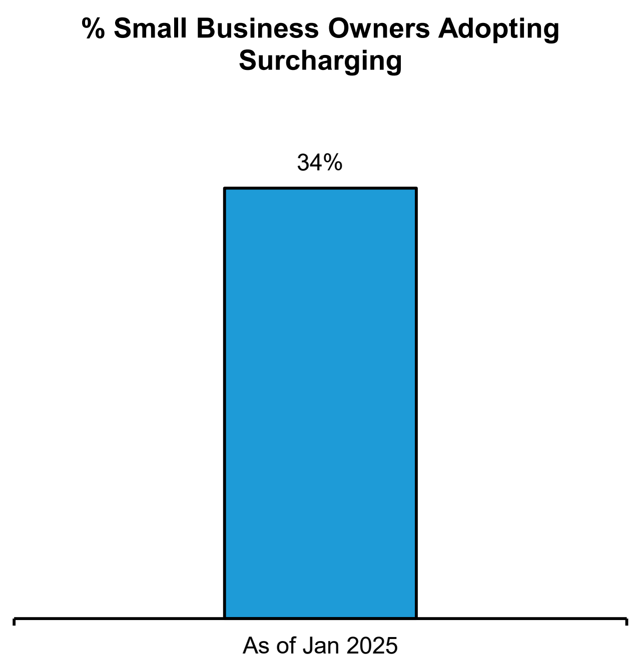
ALL THAT SAID, IN THE U.S., THE COST OF ACCEPTANCE IS AT A SOMEWHAT DELICATE MOMENT

U.S. credit card rates are higher vs. other countries, in part driven by growing share of premium credit cards which carry high interchange. An average merchant in the U.S. pays 2.4% fees, although this is a weighted average number — large merchants (e.g., Amazon, Walmart) pay significantly lower vs small merchants (who often pay north of 3% for cards). As such, **we are increasingly seeing small business owners adopting surcharging (Exhibit 8). Note that the MDL settlement (likely to be approved later this year) will further enable more flexible ways to surcharge.** See our note on MDL - Payments: Visa/Mastercard Merchant Settlement; Our Thoughts. **Note that surcharging is only allowed for credit cards, not debit cards (which are already cheaper).**

While the overall % of surcharging is currently low in the U.S., we worry about the penetration upticking higher if/when MDL settlement is approved. Merchant acquirers such as Toast have additional incentive to surcharge given the monetization model/ARPU upside. Then there are occasional reports around some merchants not taking cards e.g., Meta recently revamped its ad payment policy for some advertisers to make their monthly payments via invoice/debit (vs. cards).

While inconsequential in the near-term, we do somewhat worry about long-term consequences of detrimental user experience around surcharging and also the impact on merchant card acceptance. It is a tricky topic the networks need to address with their issuer customers (who want higher interchange) to maintain consumers' (and merchants') strong preference for cards over time. We note that Amex rules do not allow surcharging (it won that ruling in the Supreme Court in 2018).

EXHIBIT 8: **Surcharging among small business owners reached 34% per a survey conducted early 2025**



Source: JD Power, Bernstein analysis

COMMERCIAL MODELS CHANGED HISTORICALLY TO ACCOMMODATE NEWER CATEGORIES, AND THEY WILL CHANGE AGAIN IN THE FUTURE

Groceries, vending machines and transit, at different points of time, were not addressable by cards because of how the pricing worked - both on interchange and fixed/variable components of payment fees. Over time, cards changed their pricing models (and also interchange) to accommodate these categories.

We are closely watching how cost of acceptance evolves for certain categories and merchants, especially in the U.S. (where both credit card usage and fees are higher). We are monitoring two specific 'new markets':

1) B2B/Commercial — the networks already have a flexible interchange model for buyers/sellers. It's possible that B2B digitization by cards gets accelerated by AI agents, and flexible

interchange will help, among other things,

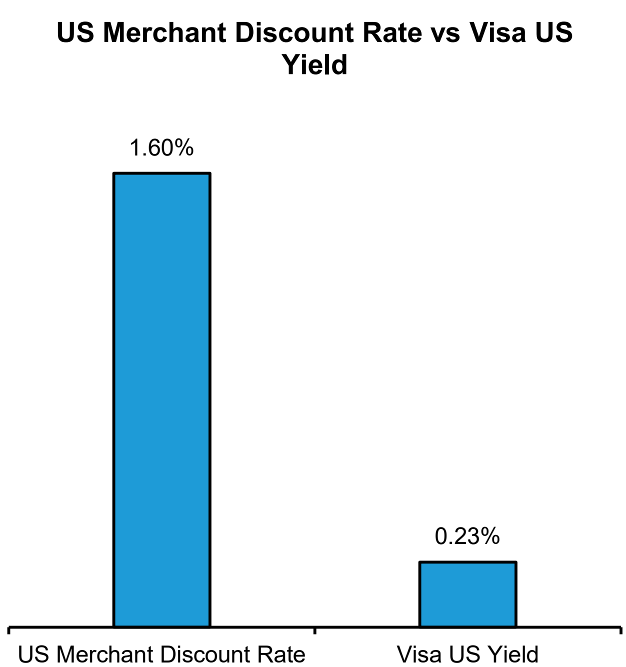
2) Autonomous agent micro-transactions — a new economic construct where the command line interface or CLI becomes a commerce platform e.g., for agents buying compute, images, tokens or interacting with APIs. Visa, for e.g., is building the ability for people to be able to load their Visa card into the CLI, and have their agents autonomously make purchases (including very small ticket transactions). As these models evolve, V/MA can be flexible with their pricing models to ensure they remain the best way to pay for small transactions.

See our research: The Age of Agents: Insights from our Inaugural Agentic Commerce Day and The Age of Agents: How does it change commerce and how we pay?

ARE NETWORKS' OVER-EARNING?

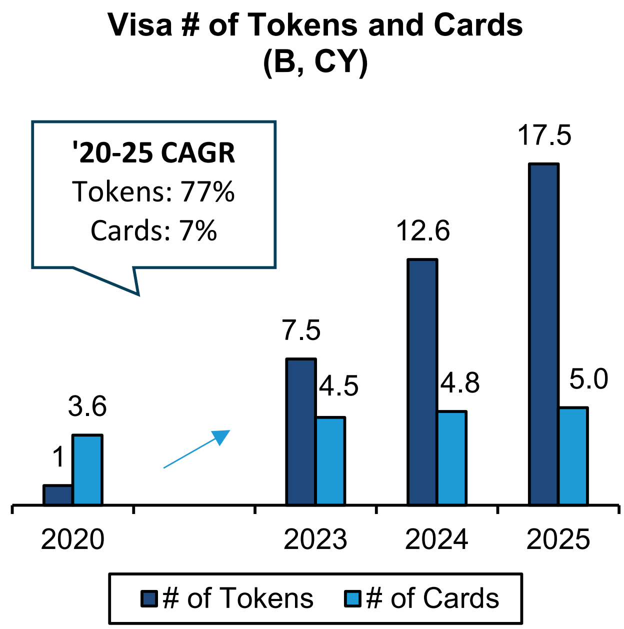
Visa and Mastercard net yields (a proxy for network fees) are a fraction of average MDR in the U.S. (Exhibit 9). The networks, in our view, are NOT over-earning. Card networks add enormous value in a transaction relative to their fees and market position (100m+ merchant acceptance points, 7B+ KYCed credentials) and likely will continue to have pricing power especially as token penetration grows (Exhibit 10) including with agentic commerce.

EXHIBIT 9: **The networks yields are only a small fraction of total MDR**



MA doesn't disclose US revenue separately
Source: Company reports, Bernstein analysis

EXHIBIT 10: **Tokens are proliferating, and will do so even more with Agentic commmerce - solidifying moat and pricing power**



Source: Company reports and Bernstein analysis

TO CONCLUDE..

Cost of acceptance of cards is a nuanced topic, especially in the U.S. Regulated debit cards are already quite cheap, and credit cards are different. We will be closely watching how things evolve (e.g., around surcharging) evolve in the U.S. post MDL settlement approval. Longer-term, the networks have shown flexibility in certain commercial models (e.g., in interchange) to grow the ecosystem, and can continue to strike the delicate balance between growing acceptance and consumer preference/experience.

PLEASE SEE LINKS TO OUR RECENT RESEARCH

- Payments: Stablecoins — What we learned from our dozens of industry conversations
- The Age of Agents: Insights from our Inaugural Agentic Commerce Day
- Visa, Mastercard: Payments-ocalypse, will AI doom the networks?
- Visa, Mastercard: Why we like the networks more now vs. 5yrs ago

Harshita Rawat, CFA +1 917 344 8485 harshita.rawat@bernsteinsg.com

BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	4 Jun 2026 Closing Price	Price Target	TTM Rel. Perf.	Adjusted EPS Cur	Adjusted EPS 2025A	Adjusted EPS 2026E	Adjusted EPS 2027E	Adjusted P/E (x) 2025A	Adjusted P/E (x) 2026E	Adjusted P/E (x) 2027E
V (Visa)	O	USD	320.18	450.00	(39.7)%	USD	11.47	13.17	15.16	27.9	24.3	21.1
MA (Mastercard)	O	USD	481.76	710.00	(44.7)%	USD	17.01	19.82	22.82	28.3	24.3	21.1
PYPL (PayPal)	M	USD	42.75	45.00	(68.0)%	USD	5.31	5.23	5.30	8.0	8.2	8.1
ADYEN.NA (Adyen)	O	EUR	897.00	1,600.00	(60.2)%	EUR	33.62	42.83	51.05	26.7	20.9	17.6
ADYEY (Adyen)	O	USD	10.36	18.63	(73.3)%	USD	0.39	0.50	0.60	26.4	20.6	17.3
FIS (FIS)	M	USD	41.48	73.00	(75.5)%	USD	5.76	6.30	6.91	7.2	6.6	6.0
FI (Fiserv)	M	USD	56.23	76.00	(93.0)%	USD	8.64	8.06	8.76	6.5	7.0	6.4
GPN (Global Payments)	M	USD	67.81	86.00	(37.8)%	USD	12.22	13.89	16.13	5.5	4.9	4.2
TOST (Toast)	O	USD	25.22	39.00	(68.1)%	USD	633.00	802.78	990.80	20.3	16.0	13.0
XYZ (Block Inc)	O	USD	70.89	85.00	(15.6)%	USD	2,084	3,347	4,700	44.2%	50.2%	32.0%
SPX			7,584.31									
EDME			1,548.21									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

TOST estimate is EBITDA (M); XYZ estimate is EBIT (M); TOST valuation is EV/EBITDA (x); XYZ valuation is EBIT CAGR;

Source: Bloomberg, Bernstein estimates and analysis.

I. REQUIRED DISCLOSURES

References to "Bernstein" or the “Firm” in these disclosures relate to the following entities: Bernstein Institutional Services LLC (April 1, 2024 onwards), Sanford C. Bernstein & Co., LLC (pre April 1, 2024), Bernstein Autonomous LLP, BSG France S.A. (April 1, 2024 onwards), Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, Sanford C. Bernstein (Canada) Limited, Sanford C. Bernstein (India) Private Limited (SEBI registration no. INH000006378), Sanford C. Bernstein (Singapore) Private Limited, Sanford C. Bernstein Japan KK（サンフォード・C・バーンスタイン株式会社）and analysts employed by Société Générale Africa Technologies & Services to produce Bernstein research under a Global Services Agreement in place between Bernstein and Société Générale.

Bernstein is part of a joint venture between Société Générale (SG) and AllianceBernstein, L.P. (AB). Unless specifically noted otherwise, for purposes of these disclosures, references to Bernstein’s “affiliates” relate to both SG and AB and their respective affiliates.

VALUATION METHODOLOGY

This research publication covers six or more companies. For valuation methodology and other company disclosures:
Please visit: https://bernstein-autonomous.bluematrix.com/sellside/Disclosures.action.
Or, you can also write to the Director of Compliance, Bernstein Institutional Services LLC, 245 Park Avenue, New York, NY 10167.

RISKS

This research publication covers six or more companies. For risks and other company disclosures:
Please visit: https://bernstein-autonomous.bluematrix.com/sellside/Disclosures.action.
Or, you can also write to the Director of Compliance, Bernstein Institutional Services LLC, 245 Park Avenue, New York, NY 10167.

RATINGS DEFINITIONS, BENCHMARKS AND DISTRIBUTION

EQUITY RATINGS DEFINITIONS

Bernstein brand

The Bernstein brand rates stocks based on forecasts of relative performance for the next 12 months versus the S&P 500 for stocks listed on the U.S. and Canadian exchanges, versus the Bloomberg Europe Developed Markets Large and Mid Cap Price Return Index EUR (EDME) for stocks listed on the European exchanges and emerging markets exchanges outside of the Asia Pacific region, versus the Bloomberg Japan Large and Mid Cap Price Return Index USD (JPL) for stocks listed on the Japanese exchanges, and versus the Bloomberg Asia ex-Japan Large and Mid Cap Price Return Index (ASIAX) for stocks listed on the Asian (ex-Japan) exchanges -unless otherwise specified.

The Bernstein brand has three categories of ratings:

- Outperform: Stock will outpace the market index by more than 15 pp
- Market-Perform: Stock will perform in line with the market index to within +/-15 pp
- Underperform: Stock will trail the performance of the market index by more than 15 pp

Coverage Suspended: Coverage of a company under the Bernstein research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Not Covered (NC) denotes companies that are not under coverage.

Bernstein brand stock ratings are based on a 12-month time horizon.

Autonomous brand – common stocks

The Autonomous brand rates common stocks as indicated below. As our benchmarks we use the Bloomberg Europe 500 Banks And Financial Services Index (BEBANKS) and Bloomberg Europe Dev Mkt Financials Large and Mid Cap Price Ret Index EUR (EDMFI) index for developed European banks and Payments, the Bloomberg Europe 500 Insurance Index (BEINSUR) for European insurers, the S&P 500 and S&P Financials for US banks and Payments coverage, S5LIFE for US Insurance, the S&P Insurance Select Industry (SPSIINS) for US Non-Life Insurers coverage, and the Bloomberg Emerging Markets Financials Large, Mid and Small Cap Price Return Index (EMLSF) for emerging market banks and insurers and Payments. Ratings are stated relative to the sector (not the market).

The Autonomous brand has three categories of common stock ratings:

- Outperform (OP): Stock will outpace the relevant index by more than 10 pp
- Neutral (N): Stock will perform in line with the market index to within +/-10 pp
- Underperform (UP): Stock will trail the performance of the relevant index by more than 10 pp

Coverage Suspended: Coverage of a company under the Autonomous research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Those denoted as 'Feature' (e.g., Feature Outperform FOP, Feature Under Outperform FUP) are our core ideas.

Not Covered (NC) denotes companies that are not under coverage.

Autonomous brand common stock ratings are based on a 12-month time horizon.

Autonomous brand – preferred stocks

The Autonomous brand has three categories of preferred stock ratings:

- Outperform (OP): The total return of the preferred instrument is expected to outperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Neutral (N): The total return of the preferred instrument is expected to perform in line with preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Underperform (UP): The total return of the preferred instrument is expected to underperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous preferred stock ratings are based on a 6-month time horizon.

AUTONOMOUS CREDIT RESEARCH

Where this report contains investment recommendations for credit instruments, as defined in article 3(1)(35) of the Market Abuse Regulation, the information below is presented to comply with its disclosure requirements.

The report may also include reference(s) to published opinions by other Autonomous or Bernstein analysts covering the equity securities of the issuer(s) referenced herein. Please note an investment recommendation for credit instruments published by the author(s) of this report may differ from the published view of the analyst covering equity securities for the issuer(s) contained in this report and vice versa.

CREDIT RATINGS DEFINITIONS

The Autonomous brand has three categories of credit ratings:

- Credit Outperform (C-OP): The total return of the Reference Credit Instrument is expected to outperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.
- Credit Neutral (C-N): The total return of the Reference Credit Instrument is expected to perform in line with the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.
- Credit Underperform (C-UP): The total return of the Reference Credit Instrument is expected to underperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous credit ratings are based on a 6-month time horizon.

A list of all investment recommendations produced by the author(s) of this report alongside credit ratings history are available upon request.

It is at the sole discretion of the Firm as to when to initiate, update and cease research coverage. The Firm has established, maintains and relies on information barriers to control the flow of information contained in one or more areas (i.e. the private side) within the Firm, and into other areas, units, groups or affiliates (i.e. public side) of the Firm

DISTRIBUTION OF EQUITY RATINGS/INVESTMENT BANKING SERVICES

Equity Rating	Market Abuse Regulation (MAR) and FINRA Rating Category	Global Rating Distribution	Investment Banking Relationships*
Outperform	BUY	51.1%	16.5%
Market-Perform (Bernstein Brand) Neutral (Autonomous Brand)	HOLD	36.3%	17.8%
Underperform	SELL	12.6%	14.9%

* These figures represent the percentage of companies within each equity rating category for which affiliates of Bernstein have provided investment banking services within the previous 12 months.
As of March 31, 2026. All figures are updated quarterly.

Prior to April 1, 2024, Sanford C. Bernstein & Co., LLC. issued the ratings and price target information in the graph(s) below for the following companies: Visa Inc, Mastercard Inc, PayPal Holdings Inc, Adyen NV, FIS, Global Payments Inc and Toast Inc.

PRICE CHARTS/ RATINGS AND PRICE TARGET HISTORY

This research publication covers six or more companies. For price chart and other company disclosures, please visit https://bernstein-autonomous.bluematrix.com/sellside/Disclosures.action or you can write to the Director of Compliance, Bernstein Institutional Services LLC, 245 Park Avenue, New York, NY 10167.

Visa Inc, Mastercard Inc, PayPal Holdings Inc, Adyen NV, FIS, Fiserv, Global Payments Inc, Toast Inc and Block Inc are covered by both the Autonomous and Bernstein brands. For the research ratings and price target history please go to https://bernstein-autonomous.bluematrix.com/sellside/Disclosures.action.

CONFLICTS OF INTEREST

All statements in this report attributable to Gartner represent Bernstein's interpretation of data, research opinion or viewpoints published as part of a syndicated subscription service by Gartner, Inc., and have not been reviewed by Gartner. Each Gartner publication speaks as of its original publication date (and not as of the date of this report). The opinions expressed in Gartner publications are not representations of fact, and are subject to change without notice.

Stacy A. Rasgon and Gautam Chhugani maintains long positions in various crypto currencies.

AB and/or its affiliates beneficially own 1% or more of a class of common equity securities of the following companies: Visa Inc and Toast Inc.

Bernstein Autonomous LLP or BSG France SA, beneficially, has either a net long or short position of 0.5% or more of the total issued share capital of a class of common equity securities of the following MiFID eligible securities: Visa Inc and Toast Inc.

SG and/or its affiliates beneficially own 0.5% or more of [the total issued share capital/any class of common equity securities] with a net [long/short] position of: Visa Inc.

SG and/or its affiliates beneficially own 1% or more of a class of common equity securities of the following companies: PayPal Holdings Inc and Global Payments Inc.

Bernstein and/or affiliates have received compensation for investment banking services in the past twelve months from Mastercard Inc.

Bernstein and/or affiliates have received compensation for non-investment banking securities-related products or services in the previous twelve months from the following clients: Mastercard Inc, Fiserv and Global Payments Inc.

An affiliate of Bernstein has received compensation for non-investment banking securities-related products or services in the previous twelve months from the following clients: Fiserv and Block Inc.

Certain affiliates of Bernstein act as market maker or liquidity provider in the equities securities of: Visa Inc, Mastercard Inc, PayPal Holdings Inc, Adyen NV, FIS, Global Payments Inc, Toast Inc and Block Inc.

Certain affiliates of Bernstein act as market maker or liquidity provider in the debt securities of: Visa Inc, Mastercard Inc and FIS.

Affiliates of Bernstein managed or co-managed in the past twelve months a public offering of securities of Mastercard Inc.

Bernstein and/or affiliates had an investment banking client relationship during the past twelve months with Mastercard Inc.

OTHER MATTERS

The legal entity(ies) employing the analyst(s) listed in this report, and their location, can be determined by the country code of their phone number, as follows:

+1 Bernstein Institutional Services LLC; New York, New York, USA

+44 Bernstein Autonomous LLP; London UK

+212 Société Générale Africa Technologies & Services; Casablanca, Morocco

+33 BSG France S.A.; Paris, France

+34 BSG France S.A.; Madrid, Spain

+41 Bernstein Autonomous LLP; Geneva, Switzerland

+49 BSG France S.A.; Frankfurt, Germany

+91 Sanford C. Bernstein (India) Private Limited; Mumbai, India

+852 Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司; Hong Kong, China

+65 Sanford C. Bernstein (Singapore) Private Limited; Singapore

+81 Sanford C. Bernstein Japan KK; Tokyo, Japan

Where this report has been prepared by research analyst(s) employed by a non-US affiliate, such analyst(s), is/are (unless otherwise expressly noted below) not registered as associated persons of Bernstein Institutional Services LLC or any other SEC-registered broker-dealer and are not licensed or qualified as research analysts with FINRA. Accordingly, such analyst(s) may not be subject to FINRA's restrictions regarding (among other things) communications by research analysts with a subject company, interactions between research analysts and investment banking personnel, participation by research analysts in solicitation and marketing activities relating to investment banking transactions, public appearances by research analysts, and trading securities held by a research analyst account.

Where this report has been prepared by research analyst(s) employed by Société Générale Africa Technologies & Services (part of the Société Générale group of companies), it has been prepared on behalf of a Bernstein company under a Global Services Agreement in place between Bernstein and Société Générale.

CERTIFICATION

Each research analyst listed in this report, who is primarily responsible for the preparation of the content of this report, certifies that all of the views expressed in this publication accurately reflect that analyst's personal views about any and all of the subject securities or issuers and that no part of that analyst's compensation was, is, or will be, directly or indirectly, related to the specific recommendations or views in this publication.

II. ADDITIONAL GLOBAL CONFLICT DISCLOSURES

It is at the sole discretion of the Firm as to when to initiate, update and cease research coverage. The Firm has established, maintains and relies on information barriers to control the flow of information contained in one or more areas (i.e., the private side) within the Firm, and into other areas, units, groups or affiliates (i.e., public side) of the Firm.

III. OTHER IMPORTANT INFORMATION AND DISCLOSURES

Separate branding is maintained for "Bernstein" and "Autonomous" research products.

- Bernstein produces a number of different types of research products including, among others, fundamental analysis and quantitative analysis under both the "Autonomous" and "Bernstein" brands. Recommendations contained within one type of research product may differ from recommendations contained within other types of research products, whether as a result of differing time horizons, methodologies or otherwise. Furthermore, views or recommendations within a research product issued under one brand may differ from views or recommendations under the same type of research product issued under the other brand. The Research Ratings System for the two brands and other information related to those Rating Systems are included in the previous section.

- Autonomous operates as a separate business unit within the following entities: Bernstein Institutional Services LLC, Bernstein Autonomous LLP, Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司 and Sanford C. Bernstein (India) Private Limited. For information relating to "Autonomous" branded products (including certain Sales materials) please visit: www.autonomous.com. For information relating to Bernstein branded products please visit: www.bernsteinresearch.com.

Analysts are compensated based on aggregate contributions to the research franchise as measured by account penetration, productivity and proactivity of investment ideas. No analysts are compensated based on performance in, or contributions to, generating investment banking revenues.

This report has been produced by an independent analyst as defined in Article 3 (1)(34)(i) of EU 596/2014 Market Abuse Regulation ("MAR") and the same article of MAR as it forms part of United Kingdom domestic law by virtue of the European Union (Withdrawal) Act 2018.

To our readers in the United States: Bernstein Institutional Services LLC, a broker-dealer registered with the U.S. Securities and Exchange Commission ("SEC") and a member of the U.S. Financial Industry Regulatory Authority, Inc. ("FINRA") is distributing this publication in the United States and accepts responsibility for its contents. Where this material contains an analysis of debt product(s), such material is intended only for institutional investors and is not subject to the US independence and disclosure standards applicable to debt research prepared for retail investors.

Bernstein Institutional Services LLC may act as principal for its own account or as agent for another person (including an affiliate) in sales or purchases of any security which is a subject of this report. This report does not purport to meet the objectives or needs of any specific individuals, entities or accounts.

To our readers in Canada: If this publication pertains to a Canadian domiciled company, it is being distributed in Canada by Sanford C. Bernstein (Canada) Limited, which is licensed and regulated by the Canadian Investment Regulatory Organization. If the publication pertains to a non-Canadian domiciled company, it is being distributed by Bernstein Institutional Services LLC, which is licensed and regulated by both the SEC and FINRA, into Canada under the International Dealers Exemption.

This document may not be passed onto any person in Canada unless that person qualifies as "permitted client" as defined in Section 1.1 of NI 31-103.

To our readers in Brazil: This report has been prepared by Bernstein Institutional Services LLC, and Banco BTG Pactual S.A. ("BTG") is responsible for the distribution of this report in Brazil.

To readers in the United Kingdom: This publication has been issued or approved for issue in the United Kingdom by Bernstein Autonomous LLP, authorised and regulated by the Financial Conduct Authority and located at 60 London Wall, London EC2M 5SH, +44 (0)20-7170-5000. Registered in England & Wales No OC343985.

This document is for distribution only to persons who (i) have professional experience in matters relating to investments falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the “Financial Promotion Order”), (ii) are persons falling within Article 49(2)(a) to (d) (“high net worth companies, unincorporated associations, etc.”) of the Financial Promotion Order, (iii) are outside the United Kingdom, or (iv) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the FSMA) in connection with the issue or sale of any securities may otherwise lawfully be communicated or caused to be communicated (all such persons together being referred to as “relevant persons”). This document is directed only at relevant persons and must not be acted on or relied on by persons who are not relevant persons. Any investment or investment activity to which this document relates is available only to relevant persons and will be engaged in only with relevant persons.

To our readers in the member states of the EEA: This publication is being distributed by BSG France SA, which is authorised and regulated by the Autorité de Contrôle Prudentiel et de Résolution (ACPR) and Autorité des Marchés Financiers (AMF).

To our readers in Hong Kong: This publication is being distributed in Hong Kong by Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, which is licensed and regulated by the Hong Kong Securities and Futures Commission (Central Entity No. AXC846) to carry out Type 4 (Advising on Securities) regulated activities and subject to the licensing conditions mentioned in the SFC Public Register (https://www.sfc.hk/publicregWeb/corp/AXC846/details)). This publication is solely for professional investors, as defined in the Securities and Futures Ordinance (Cap. 571). The purpose of this report is solely to provide an analysis of the issuers referred to in this report and is not intended for any purpose contrary to the laws of Hong Kong.

To our readers in Singapore: This publication is being distributed in Singapore by Sanford C. Bernstein (Singapore) Private Limited, only to accredited investors or institutional investors, as defined in the Securities and Futures Act 2001 of Singapore ("SFA"). Recipients in Singapore should contact Sanford C. Bernstein (Singapore) Private Limited in respect of matters arising from, or in connection with, this publication. Sanford C. Bernstein (Singapore) Private Limited is regulated by the Monetary Authority of Singapore and licensed under the SFA as a capital markets services licence holder for dealing in capital markets products that are securities and collective investment schemes and an exempt financial adviser for advising on, issuing and promulgating analyses and reports on securities. Sanford C. Bernstein (Singapore) Private Limited is registered in Singapore with Company Registration No. 20213710W and located at One Raffles Quay, #27-11 South Tower, Singapore 048583, +65-6230-4612.

To our readers in the People’s Republic of China: The securities referred to in this document are not being offered or sold and may not be offered or sold, directly or indirectly, in the People's Republic of China (for such purposes, not including the Hong Kong and Macau Special Administrative Regions or Taiwan, the “PRC”) in contravention of any applicable laws of the PRC.

This document does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC to any person to whom it is unlawful to make the offer or solicitation in the PRC.

We do not represent that this document may be lawfully distributed, or that any securities may be lawfully offered, in compliance with any applicable registration or other requirements in the PRC, or pursuant to an exemption available thereunder, or assume any responsibility for facilitating any such distribution or offering. In particular, no action has been taken by us which would permit a public offering of any securities or distribution of this document in the PRC. Accordingly, the securities are not being offered or sold within the PRC by means of this document or any other document. Neither this document nor any advertisement or other offering material may be distributed or published in the PRC, except under circumstances that will result in compliance with any applicable laws and regulations.

To our readers in Japan: This publication is being distributed in Japan by Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社), which is registered in Japan as a Financial Instruments Business Operator with the Kanto Local Finance Bureau (registration number: The Director-General of Kanto Local Finance Bureau (FIBO) No.3387) and regulated by the Financial Services Agency. It is also a member of Japan Investment Advisers Association. This publication is solely for qualified institutional investors in Japan only, as defined in Article 2, paragraph (3), items (i) of the Financial Instruments and Exchange Act.

For the institutional client readers in Japan who have been granted access to the Bernstein website by Daiwa Securities Group Inc. ("Daiwa"), your access to this document should not be construed as meaning that Bernstein is providing you with investment advice for any purposes. Whilst Bernstein has prepared this document, your relationship is, and will remain with, Daiwa, and Bernstein has neither any contractual relationship with you nor any obligations towards you.

To our readers in Australia: Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司 is responsible for distributing research in Australia. It is regulated by the Securities and Exchange Commission under U.S. laws, by the Financial Conduct Authority under U.K. laws, which differs from Australian laws. Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司 is exempt from the requirement to hold an Australian financial services license under the Corporations Act 2001 in respect of the provision of the following financial services to wholesale clients:

- providing financial product advice;
- dealing in a financial product;
- making a market for a financial product; and
- providing a custodial or depository service.

To our readers in India: This publication is being distributed in India by Sanford C. Bernstein (India) Private Limited (SCB India) which is licensed and regulated by Securities and Exchange Board of India ("SEBI") as a research analyst entity under the SEBI (Research Analyst) Regulations, 2014, having registration no. INH000006378 and as a stock broker having registration no. INZ000213537. SCB India is currently engaged in the business of providing research and stock broking services. Please refer to www.bernsteinresearch.in for more information.

- SCB India is a Private limited company incorporated under the Companies Act, 2013, on April 12, 2017 bearing corporate identification number U65999MH2017FTC293762, and registered office at Level 3A, 4th Floor, First International Financial Centre, Plot Nos C-54 and C-55, G Block, Near CBI Office, Bandra Kurla Complex, Bandra (East), Mumbai 400098, Maharashtra, India (Phone No: +91-22-68421401).
- For details of Associates (i.e., affiliates/group companies) of SCB India, kindly email MUM-BERNSTEIN-InCompliance@bernsteinsg.com.
- SCB India does not have any disciplinary history as on the date of this report.
- Except as noted above, SCB India and/or its Associates (i.e., affiliates/group companies), the Research Analysts authoring this report, and their relatives
    - do not have any financial interest in the subject company
    - do not have actual/beneficial ownership of one percent or more in securities of the subject company;
    - is not engaged in any investment banking activities for Indian companies, as such;
    - have not managed or co-managed a public offering in the past twelve months for any Indian companies;
    - have not received any compensation for investment banking services or merchant banking services from the subject company in the past 12 months;
    - have not received compensation for brokerage services from the subject company in the past twelve months;

- have not received any compensation or other benefits from the subject company or third party related to the specific recommendations or views in this report; and

- do not currently, but may in the future, act as a market maker in the financial instruments of the companies covered in the report.

- do not have any conflict of interest in the subject company as of the date of this report.

- Except as noted above, the subject company has not been a client of SCB India during twelve months preceding the date of distribution of this research report. Neither SCB India nor its Associates (i.e., affiliates/group companies) have received compensation for products or services other than investment banking, merchant banking or brokerage services from the subject company in the past twelve months.

- The principal research analyst(s) who prepared this report, members of the analysts' team, and members of their households are not an officer, director, employee or advisory board member of the companies covered in the report.

- Our Compliance officer / Grievance officer is Ms. Rupal Talati, who can be reached at +91-22-68421451, or MUM-BERNSTEIN-InCompliance@bernsteinsg.com / Scbin-investorgrievance@bernsteinsg.com

- The Research investor charter and Terms & Conditions of SCB India are available on its website and may be accessed at Sanford C. Bernstein (India) Private Limited (https://bernsteinresearch.in/) for your reference.

- Disclaimer: Registration granted by SEBI, and certification from NISM, is in no way a guarantee of performance of the intermediary or provide any assurance of returns to investors. Investments in securities market are subject to market risks. Read all the related documents carefully before investing.

To our readers in Switzerland: This document is provided in Switzerland by or through Bernstein Autonomous LLP, and is provided only to qualified investors as defined in article 10 of the Swiss Collective Investment Scheme Act (“CISA”) and related provisions of the Collective Investment Scheme Ordinance and in strict compliance with applicable Swiss law and regulations. The products mentioned in this document may not be suitable for all types of investors. This document is based on the Directives on the Independence of Financial Research issued by the Swiss Bankers Association (SBA) in January 2008.

To our readers in the Middle East: Bernstein Autonomous LLP, DIFC branch has its principal office at Gate Village 06, DIFC, Dubai, UAE. Bernstein Autonomous LLP, DIFC branch is regulated by the Dubai Financial Services Authority (DFSA) with the registration number CL10040 and is provisioned for Arranging Deals in Investments and Advising on Financial Products. All communications and services are directed at Professional Clients and Market Counterparties only (as defined in the DFSA rulebook). Persons other than Professional Clients and Market Counterparties, such as Retail Clients, are not the intended recipients of our communications or services.

LEGAL

All research publications are disseminated to our clients through posting on the firm's password protected websites, bernsteinresearch.com and autonomous.com. Certain, but not all, research publications are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience.

This publication has been published and distributed in accordance with the Firm's policy for management of conflicts of interest in investment research, a copy of which is available from Bernstein Institutional Services LLC, Director of Compliance, 245 Park Avenue, New York, NY 10167. Additional disclosures and information regarding Bernstein's business are available on our website www.bernsteinresearch.com.

The securities described herein may not be eligible for sale in all jurisdictions or to certain categories of investors where that permission profile is not consistent with the licenses held by the entities noted herein. This document is for distribution only as may be permitted by law. This publication is not directed to, or intended for distribution to or use by, any person or entity who is a citizen or resident of, or located in any locality, state, country or other jurisdiction where such distribution, publication, availability or use would be contrary to law or regulation or which would subject any of the entities referenced herein or any of their subsidiaries or affiliates to any registration or licensing requirement within such jurisdiction. This publication is based upon public sources we believe to be reliable, but no representation is made by us that the publication is accurate or complete. We do not undertake to advise you of any change in the reported information or in the opinions herein. This publication was prepared and issued by entity referred to herein for distribution to eligible counterparties or professional clients. This publication is not an offer to buy or sell any security, and it does not constitute investment, legal or tax advice. The investments referred to herein may not be suitable for you. Investors must make their own investment decisions in consultation with their professional advisors in light of their specific circumstances. The value of investments may fluctuate, and investments that are denominated in foreign currencies may fluctuate in value as a result of exposure to exchange rate movements. Information about past performance of an investment is not necessarily a guide to, indicator of, or assurance of, future performance.

This report is directed to and intended only for our clients who are “eligible counterparties”, “professional clients”, “institutional investors” and/or “professional investors” as defined by the aforementioned regulators, and must not be redistributed to retail clients as defined by the aforementioned regulators. Retail clients who receive this report should note that the services of the entities noted herein are not available to them and should not rely on the material herein to make an investment decision. The result of such act will not hold the entities noted herein liable for any loss thus incurred as the entities noted herein are not registered/ authorised/ licensed to deal with retail clients and will not enter into any contractual agreement/arrangement with retail clients. This report is provided subject to the terms and conditions of any agreement that the clients may have entered into with the entities noted herein. All research reports are disseminated on a simultaneous basis to eligible clients through electronic publication to our client portal.

The information in this report was prepared by Bernstein solely for the internal business use of our clients. Clients may store, display, analyze, reformat and print the information in this report for this limited use only. Clients may not copy, alter, create derivative works, resell, reverse engineer, commercially exploit, share or distribute any part of the information contained herein for any purpose without Bernstein’s express written consent. These restrictions include extracting data or using the content to develop indices or other products. Further, you may not use this report, or any portion of this report, to train or finetune any third-party machine learning or artificial intelligence system, or as a prompt or input into any such system. You also may not, without Bernstein’s express written consent, do any of the foregoing in connection with your own internal machine learning or artificial intelligence system.

Bernstein may use artificial intelligence tools in the preparation of its materials. Any such materials are reviewed by Bernstein’s research analysts prior to publication.

This report has been prepared for information purposes only and is based on current public information that we consider reliable, but the entities noted herein do not warrant or represent (express or implied) as to the sources of information or data contained herein are accurate, complete, not misleading or as to its fitness for the purpose intended even though the entities noted herein rely on reputable or trustworthy data providers, it should not be relied upon as such. Opinions expressed are the author(s)’ current opinions as of the date appearing on the material only and we do not undertake to advise you of any change in the reported information or in the opinions herein.

Any references to Societe Generale herein are purely factual, based upon publicly available information, and included for comparative purposes only. They do not constitute an opinion or recommendation with respect to the securities of Societe Generale.

This publication was prepared and issued by the entity referred to herein for distribution to eligible counterparties or professional clients. The information in this report is intended for general circulation and does not constitute an offer to buy or sell any security, investment, legal or tax advice nor a personal recommendation, as defined by any of the aforementioned regulators. It does not take into account the particular investment objectives, financial situations, or needs of individual investors. The report has not been reviewed by any of the aforementioned regulators and does not represent any official recommendation from the aforementioned regulators. The investments referred to herein may not be suitable for you. Investors must make their own investment decisions in consultation with advice sought from a financial adviser regarding the suitability of the investment product, taking into account the specific investment objectives, financial situation or particular needs of any recipient of the recommendation, before the recipient makes a commitment to purchase the investment product.

The analysis contained herein is based on numerous assumptions. Different assumptions could result in materially different results. The information in this report does not constitute, or form part of, any offer to sell or issue, or any offer to purchase or subscribe for shares, or to induce engage in any other investment activity. The value of any securities or financial instruments mentioned in this report may fluctuate subject to market conditions. Information about past performance of an investment is not necessarily a guide to, indicative of, or assurance of future performance. Estimates of future performance mentioned by the research analyst in this report are based on assumptions that may not be realized due to unforeseen factors like market volatility/fluctuation. In relation to securities or financial instruments denominated in a foreign currency other than the clients’ home currency, movements in exchange rates will have an effect on the value, either favorable or unfavorable. Before acting on any recommendations in this report, recipients should consider the appropriateness of investing in the subject securities or financial instruments mentioned in this report and, if necessary, seek for independent professional advice.

The securities described herein may not be eligible for sale in all jurisdictions or to certain categories of investors where that permission profile is not consistent with the licenses held by the entities noted herein. This document is for distribution only as may be permitted by law. It is not directed to, or intended for distribution to or use by, any person or entity who is a citizen or resident of or located in any locality, state, country or other jurisdiction where such distribution, publication, availability or use would be contrary to law or regulation or would subject the entities noted herein to any regulation or licensing requirement within such jurisdiction.

Source: Bloomberg Index Services Limited. BLOOMBERG® is a trademark and service mark of Bloomberg Finance L.P. and its affiliates (collectively "Bloomberg"). Bloomberg or Bloomberg's licensors own all proprietary rights in the Bloomberg Indices. Neither Bloomberg nor Bloomberg's licensors approves or endorses this material, or guarantees the accuracy or completeness of any information herein, or makes any warranty, express or implied, as to the results to be obtained therefrom and, to the maximum extent allowed by law, neither shall have any liability or responsibility for injury or damages arising in connection therewith.